

Word Order Variation in Spoken and Written Corpora

A Cross-Linguistic Study of SVO and Alternative Orders

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Introduction

- Word order is central in linguistic typology (WALS, Grambank).
- Most typological generalisations come from written data, missing modality effects.
- Recent corpus studies show word order is **gradient** and **context-sensitive**.
- Research gap:** lack of systematic cross-linguistic study of modality effects.

Objective

- Compare **spoken** vs. **written word order** across **five Indo-European languages** (English, French, Norwegian Nynorsk, Slovenian, Spanish).
- RQ1:** Does word order differ between spoken and written language?
- RQ2:** If so, how does it differ?

Data

- Spoken/written UD treebanks (v2.15), matched per language.
- Corpora: **GUM**, **Rhapsodie/GSD**, **NynorskLIA/Nynorsk**, **SST/SSJ**, **COSER/GSD**.

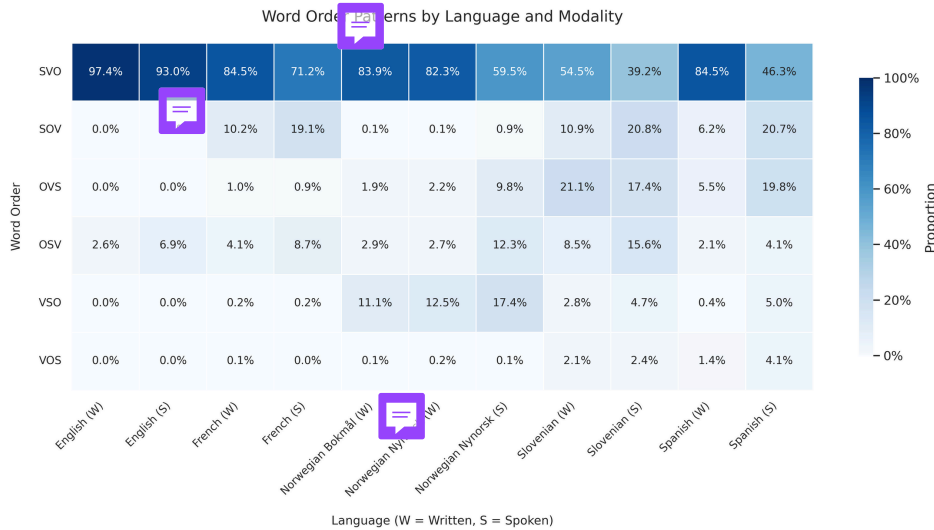
Methodology

- Corpus preparation:**
 - Excluded mixed-genre or code-switching corpora.
 - For English, the GUM treebank was manually split into spoken and written subsets based on genre metadata.
- Data extraction:** Used **STARK** to query UD-formatted dependency trees.
 - Retrieved all clauses with finite verb + nominal subject + direct object, regardless of clause type.
- Classification:** Assigned each clause to one of six canonical word orders: SVO, SOV, VSO, VOS, OSV, OVS.
- Analysis:** Compared order distributions across spoken and written modalities for each language.

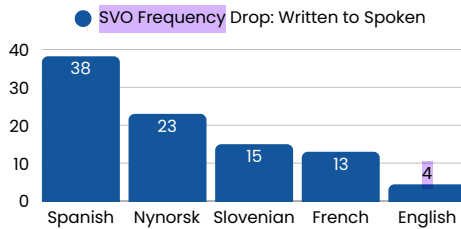
Order	Slovenian	Gloss
SVO	Mama kupuje jabolka	mother buys apples
SOV	Mama jabolka kupuje	mother apples buys
VSO	Kupuje mama jabolka	buys mother apples
VOS	Kupuje jabolka mama	buys apples mother
OSV	Jabolka mama kupuje	apples mother buys
OVS	Jabolka kupuje mama	apples buys mother

Table 1: Canonical word order examples in Slovenian.
All sentences translate as 'Mother buys apples'.

Results and Analysis



- Spoken** shows **more variation** in all languages.
- SVO** frequency **drops in speech**; SOV, OSV, OVS rise.
- Biggest spoken shift: Slovenian (SVO only 39.2%) and Spanish (SVO 46.3%).
- Smallest shift: English (SVO ~93% spoken).
- Spoken variation linked to:
 - Morphological richness (Slovenian, Spanish).
 - Prosody and discourse structure (topic/focus).
 - Cognitive pressures in real-time speech.
- Frequent non-canonical orders in speech: **object-fronting**, **postverbal subjects**.
- Cross-linguistic trend: **speech** enables **pragmatic reordering** more than writing.



- Limitations:**
 - Five IE languages
 - Uneven corpus size/genres
 - Overt S + O only
 - Verbal predicates only
 - Annotation inconsistencies

Conclusion

- Modality strongly affects word order flexibility.
- Written-based** typologies underestimate spoken variation.
- Suggests integrating spoken corpus data into typological databases.
- Future work: more languages, **genre** distinctions, entropy metrics.